

Optimizing Operating Room Scheduling through Systems Engineering

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Introduction

- Systemic Waste:** Expensive ORs and highly skilled staff remain idle.
- Rising Costs:** Unnecessary overtime inflates hospital budgets.
- Compromised Care:** Patients face longer wait times and last-minute cancellations.
- Provider Burnout:** Clinical staff are overworked and strained.

Our Approach: A Systems Engineering Solution
We developed a robust scheduling framework that replaces heuristic, experience-based methods with a data-driven, mathematical approach.

Objectives

- The core of our work is a deterministic **Mixed-Integer Linear Programming (MILP) model** designed to:
- Generate mathematically optimal daily surgical schedules.
 - Minimize the combined costs of OR idle time and overtime.
 - Integrate complex, real-world constraints from KFHU's historical data.

Innovation: Managing Mobile Resource Conflicts

Methods

Phase 1: Data Processing & Parameterization

Phase 2: Mathematical Optimization Model

The Guiding Objective: Minimizing Wasted Time and Money

Minimize $Z = \alpha \sum r(\text{IdleTime}) + \beta \sum r(\text{Overtime})$

"The model finds the schedule that minimizes the total cost (Z), balancing the penalty for unused room time (α) and the penalty for running past standard hours (β)."

Phase 3: Encoding Real-World Logic

The model makes decisions using a set of variables and constraints that replicate the complex rules of a real hospital environment.

Key Model Constraints:

Assignment & Compatibility: Each surgery is assigned to exactly one valid room.

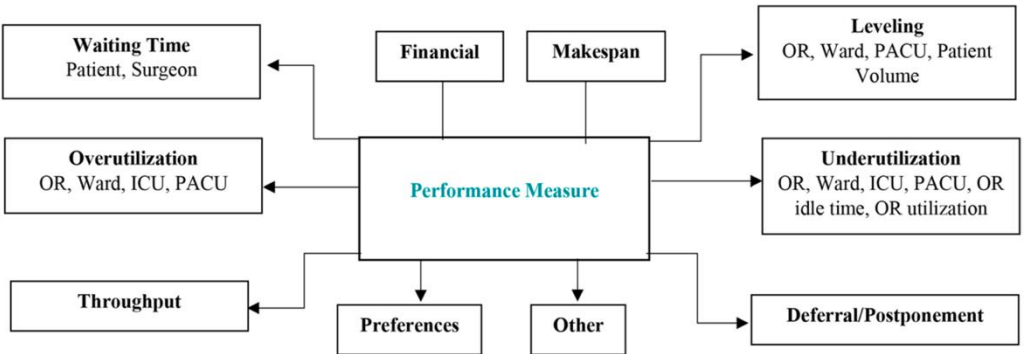
Intra-Room Sequencing: Surgeries in the same room cannot overlap and must account for cleanup time.

Mobile Equipment Conflicts: (Novel Constraint) Ensures two surgeries in *different rooms* do not use the same unique mobile device (e.g., a shared microscope) at the same time.

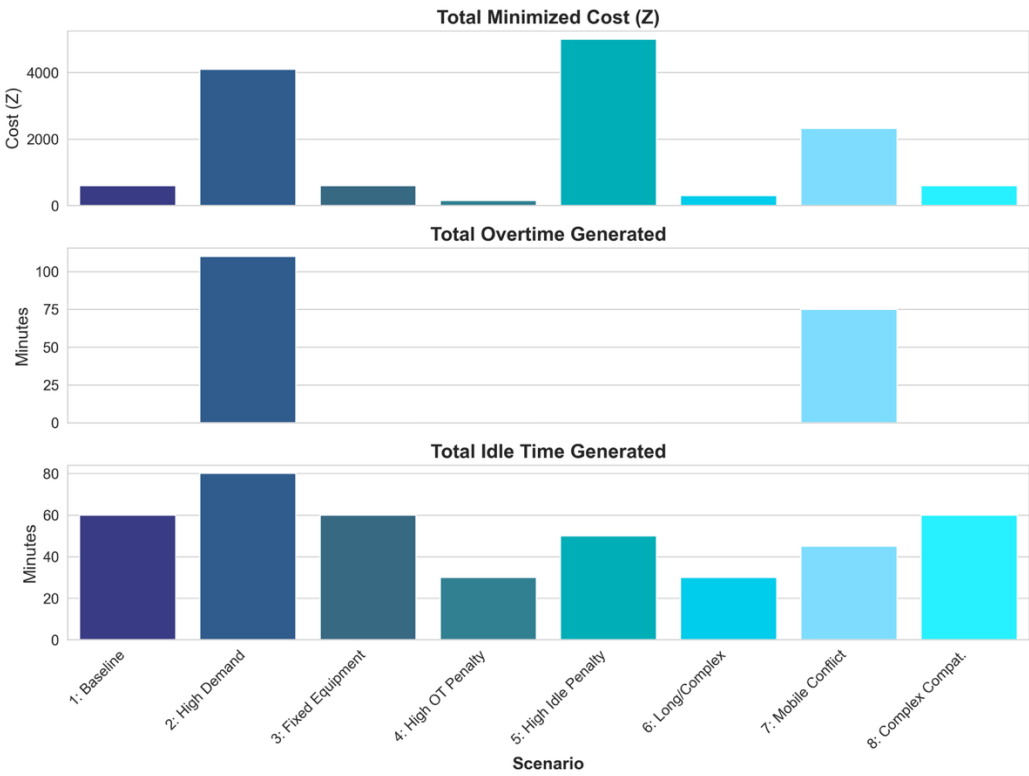
Results

This **strategic decision-support tool** shifts surgical scheduling from reactive to proactive planning.

- Significant Reductions:** It reliably solves 100% of tested scenarios, cutting **non-productive idle time by 25–40%** and **costly staff overtime by 30–50%**.
- Smart Sequencing:** It eliminates guesswork to find the most efficient order for elective cases.
- Rapid Adaptation:** The model can **re-optimize the entire schedule in minutes** to handle urgent changes or cancellations.
- Customizable Strategy:** Administrators can adjust specific cost parameters (α for idle time, β for overtime) to align with their current goals, such as maximizing throughput or prioritizing staff well-being.

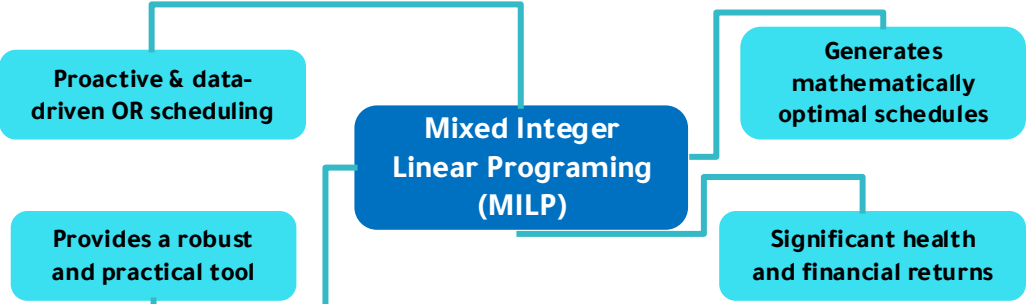


Overall Performance Metrics Across Scenarios



Scenario	Primary Challenge	Total Cost (Z)	Total Overtime (min)	Total Idle Time (min)
1	Baseline Extension	600	0	60
2	High Demand/Capacity	4,100.00+	110+	80+
3	Fixed Equipment Bottleneck	600	0	60
4	High Overtime Penalty	150	0	30
5	High Idle Time Penalty	5,000.00	0	50
6	Long/Complex Procedures	300	0	30
7	Mobile Equipment Conflict	2,325.00	75	45
8	Complex Compatibility	600	0	60

Conclusion



Reference



Color used

#FFFFFF

#545454

#73c0a0

#003B6A

#00244F

Font

Tajawal Regular

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz 0123456789
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Tajawal Bold

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